

Scalability & Future Plan

Date	15 March 2024
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform (FinRelief AI)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite database supports limited concurrent users	Reduced scalability for large user bases	High
2	AI responses depend on Google Gemini API availability and internet connectivity	Delays or service interruptions may affect recommendations	High
3	Application is deployed for local development only	Not accessible to remote users	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of concurrent users	Deploy on cloud infrastructure with load balancing and auto-scaling
2	Data Storage	SQLite database	Migrate to PostgreSQL or MySQL for large-scale deployment
3	Performance	Local FastAPI server	Implement caching, API optimization, and asynchronous processing
4	Security	Basic authentication and API protection	Add JWT authentication, HTTPS, role-based access control, and stronger encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Loan repayment tracking, reminders, and notification system	Next 3 months	Improves debt repayment management
Phase 3	Multi-language support and personalized AI financial assistant	Next 6 months	Enhances accessibility and user experience
Phase 4	Cloud deployment, banking API integration, and advanced analytics dashboard	Next 12 months	Improves scalability, automation, and real-world usability